New jobs?

The loss of many traditional jobs in everything from art to healthcare will partly be offset by the creation of new human jobs. GPs who focus on diagnosing known diseases and administering familiar treatments will probably be replaced by AI doctors. But precisely because of that, there will be much more money to pay human doctors and lab assistants to do groundbreaking research and develop new medicines or surgical procedures.[12]

AI might help create new human jobs in another way. Instead of humans competing with AI, they could focus on servicing and leveraging AI. For example, the replacement of human pilots by drones has eliminated some jobs but created many new opportunities in maintenance, remote control, data analysis and cyber security. The US armed forces need thirty people to operate every unmanned Predator or Reaper drone flying over Syria, while analysing the resulting harvest of information occupies at least eighty people more. In 2015 the US Air Force lacked sufficient trained humans to fill all these positions, and therefore faced an ironic crisis in manning its unmanned aircraft.[13]

If so, the job market of 2050 might well be characterised by human–AI cooperation rather than competition. In fields ranging from policing to banking, teams of humans-plus-AIs could outperform both humans and computers. After IBM's chess program Deep Blue beat Garry Kasparov in 1997, humans did not stop playing chess. Rather, thanks to AI trainers human chess masters became better than ever, and at least for a while human–AI teams known as 'centaurs' outperformed both humans and computers in chess. AI might similarly help groom the best detectives, bankers and soldiers in history.[14]

The problem with all such new jobs, however, is that they will probably demand high levels of expertise, and will therefore not solve the problems of unemployed unskilled labourers. Creating new human jobs might prove easier than retraining humans to actually fill these jobs. During previous waves of automation, people could usually switch from one routine low-skill job to another. In 1920 a farm worker laid off due to the mechanisation of agriculture could find a new job in a factory producing tractors. In 1980 an unemployed factory worker could start working as a cashier in a supermarket. Such occupational changes were feasible, because the move from the farm to the factory and from the factory to the supermarket required only limited retraining.

But in 2050, a cashier or textile worker losing their job to a robot will hardly be able to start working as a cancer researcher, as a drone operator, or as part of a human–AI banking team. They will not have the necessary skills.